EndoTrack

Endoso Operations & Customer Tracking Platform

Software Requirements & Product Specification

Baseline draft v0.1 — 15 October 2026

Contents

1 Document Metadata and Executive Summary 3
1.1 Document metadata . . . 3
1.2 Conventions used in this document . . . 3
1.3 Glossary . . . 3
1.4 Executive summary . . . 4

2 Business Context, Problem Statement, Objectives and Scope 4
2.1 Business context . . . 4
2.2 Problem statement . . . 5
2.3 Objectives and candidate success measures . . . 5
2.4 Scope . . . 5
2.5 Assumptions and constraints . . . 6

3 Core Domain Entities and Historical Snapshot Rules 6
3.1 Entity overview . . . 6
3.2 Customer categories and identification . . . 7
3.3 Sale register rules . . . 7
3.4 Historical snapshot rules . . . 8
3.5 Record provenance . . . 8

4 The 13-Task Operational Lifecycle 8
4.1 Status model and strict linear transition rules . . . 8
4.2 Task data and editing rules . . . 9
4.3 Financial reconciliation rules . . . 9
4.4 Task summary . . . 10
4.5 Task specifications . . . 10
4.6 Validation rules . . . 13

5 Customer Experience and the Six Macro-Milestones 13
5.1 Customer portal capabilities . . . 13
5.2 Customer account lifecycle . . . 13

5.3 Milestones . . . 14
5.4 Derived milestone state . . . 14
5.5 Customer-visible data . . . 14
5.6 Customer email notifications . . . 15

6 Security, Authentication and Role-Based Access Control 15
6.1 Actors . . . 15
6.2 Role-permission matrix . . . 16
6.3 Authentication . . . 17
6.4 Security controls . . . 18

7 Data Provenance, Event Auditing and Notification Engine Rules 18
7.1 Append-only audit trail . . . 18
7.2 Notification engine rules . . . 19
7.3 Task deadlines, reminders and escalation . . . 19
7.4 File and document management . . . 20

8 Proposed System Architecture and Technology Selection 20
8.1 Architectural principles . . . 20
8.2 Logical architecture . . . 20
8.3 Technology selection under the zero-cost constraint . . . 20
8.4 Modeling task-specific data . . . 21
8.5 Backend and frontend structure . . . 22
8.6 Key technical risks . . . 22

A Open Decisions Register 23

B Revision History 23

1 Document Metadata and Executive Summary

1.1 Document metadata

Field	Value
Document title	Software Requirements & Product Specification
Product (working name)	EndoTrack — Endoso Operations & Customer Tracking Platform **[Proposal]** (name pending approval)
Client / business owner	Grupo Pana (official Toyota and Hino dealership, Peru)
Product owner	Luis Alejandro
Prepared with	Claude (AI collaborator for product analysis and architecture)
Version / status	0.1 — baseline draft for review; not yet approved
Date	15 October 2026
Source material	`Grupo_Pana_Endoso_Master_Prompt.md` (product brief, 28 sections)
Project phase	Design (no application code has been written)

1.2 Conventions used in this document

Every statement is classified so that assumptions are never silently turned into requirements:

Tag	Meaning
[Confirmed]	Explicitly specified in the product brief; treated as a requirement or decision.
[Proposal]	A proposal or technical recommendation to be evaluated; not yet a business requirement.
[Assumption]	A temporary assumption made to progress; must be validated.
[Open]	An open question requiring a business decision or more information (see Appendix A).
CV	*Customer-visible*: the field or document must be accessible in the customer portal, subject to server-side authorization.

Untagged descriptive text summarizes confirmed material. Spanish terms used by the business (for example *endoso*, *Debida Diligencia*, *Vigencia de Poder*) are kept in their original form.

1.3 Glossary

Term	Definition
Endoso	The Grupo Pana vehicle-sales channel managed by this platform.
Sale register	The record of exactly one vehicle sale, with its own 13 tasks, documents, timeline, notes and overall state.
DDC	*Debida Diligencia del Cliente*: customer due diligence (regime `General` or `Reforzado`).
PEP	Politically Exposed Person (statuses: `No PEP`, `PEP`, `Relacionado con PEP`).
DAM	Customs Declaration of Goods.
TIVe	Electronic Vehicle Identification Card.

Term	Definition
VIN	Vehicle Identification Number.
PDS	Pre-delivery inspection center.
DNI, CE, PAS, CDI, RUC	Identification document types used for customers and employees (RUC is used by B2B customers only; CDI is a customer/signer document type).

1.4 Executive summary

Grupo Pana sells vehicles through a channel called *endoso*. Each sale passes through commercial, administrative, financial, documentation, customs, registration, preparation and delivery activities that require coordination between the customer and several employees. EndoTrack is a web platform that centralizes each operation from the official quotation to vehicle delivery.

The platform has two connected experiences:

1. An **employee application** for managing customers, vehicles, sale registers, the 13-task process, documents, assignments, notes, reminders, audit history, exports and operational analytics.
2. A **customer portal** where customers securely follow their own purchase through six customer-friendly milestones and access only the information and documents explicitly made available to them.

Key characteristics of the product:

- Each sale register represents exactly one vehicle; a customer buying two vehicles has two independent sale registers.
- The internal process is a **strictly linear sequence of 13 tasks**; the customer sees a derived six-milestone view of the same sequence.
- Customer and vehicle-price snapshots are captured at sale creation and remain immutable; significant changes are recorded in an append-only audit trail.
- Authorization, task sequencing and document visibility are enforced by the backend.
- The prototype must run at zero cost, so every component must offer a functional free tier.

The existing internal tools (a Power Apps application and a Power BI dashboard) are background context only; this document specifies the product described in the brief and does not aim to recreate them.

2 Business Context, Problem Statement, Objectives and Scope

2.1 Business context

Grupo Pana is a Peruvian automotive company and an official Toyota and Hino dealership that provides vehicle sales and other automotive services. A customer typically first contacts a sales advisor by WhatsApp or telephone and receives an initial, non-final quotation outside the tracked operation. If the customer decides to proceed, the advisor creates the sale register and the formal process starts with Task 1, the official quotation. The initial informal quotation is not a task and is outside the system. **[Confirmed]**

2.2 Problem statement

Endoso operations require coordination among customers and employees across many steps, and the current tooling is basic. The brief lists the following *potential* problem areas: operational coordination, traceability, delays, bottlenecks, document management, customer uncertainty, data quality and management reporting. **[Confirmed]** These are hypotheses to be validated during analysis rather than proven problems. **[Assumption]**

2.3 Objectives and candidate success measures

The brief lists the objectives below as initial candidates; they are not yet approved success metrics. The measures and baselines are proposals. **[Proposal]**

Objective	Candidate measure	Baseline data needed
Centralize sale information and documents	Share of active sales with all required documents stored in the platform	Current document locations and completeness audit
Improve visibility of task status, active task, order and delays	Time to answer "where is this sale?"; number of overdue tasks surfaced vs. discovered late	Current lookup effort; historical delay records
Give customers a secure, understandable view	Share of customers using the portal; inbound status-inquiry volume	Current inquiry volume per sale
Improve accountability and auditability	Share of significant changes with actor, timestamp and reason	Current traceability gaps
Identify tasks and operations needing attention	Overdue tasks per week; median overdue time	Historical task durations
Provide management analytics	Dashboard usage by managers; decisions supported	Current reporting process and cadence
Reduce avoidable communication effort	Manual status messages per sale	Current messages per sale
Protect customer, employee, financial and legal information	Zero cross-customer access incidents; security test pass rate	Security review of current practice

2.4 Scope

The following scope split is a proposal for review. **[Proposal]**

Level	Content
MVP	Employee and customer authentication; employees, customers and vehicle catalog; sale registers with snapshots; the 13-task linear workflow with required fields, uniqueness and documents; completed-task rules and reopening; cancellation; audit trail; customer portal with six milestones and customer-visible data/documents behind server-side authorization; account-creation, sale-created and quotation emails.

Level	Content
V1	Remaining customer notifications; scheduler with reminders, escalations, deduplication and notification logs; BI dashboard with role-based views; exports; DNI/RUC name lookup with manual fallback; hardening, backup/recovery and observability.
Future enhancements	Ideas that should not complicate the first release, to be evaluated later (for example additional notification channels, further integrations, or additional analytics).
Out of scope (non-goals)	Recreating the Power Apps application or Power BI dashboard; the informal initial quotation; parallel or branching tasks; customers editing their own identity or contact data; accounting or revenue recognition (the platform records amounts but cannot verify accounting facts); processing payments (payments are only recorded and verified).

2.5 Assumptions and constraints

- **[Confirmed]** The prototype must have zero hosting cost; every technology must offer a robust free tier, verified against current provider quotas.
- **[Confirmed]** Amounts are recorded in USD.
- **[Assumption]** Timestamps are stored in UTC and displayed in the Lima time zone.
- **[Open]** User-interface language(s) for employees and customers; customer emails are specified with Spanish subjects.
- **[Open]** Expected workload (sales per month, number of employees, documents per sale), needed to turn performance targets from proposals into requirements.

3 Core Domain Entities and Historical Snapshot Rules

The entities below are *business information that must be represented*. They are not a mandate for one table per row; the final relational structure will be justified during data modeling. **[Confirmed]**

3.1 Entity overview

Entity	Description and key information
Customer	Customer type; document type and number; first/last name (B2C) or legal name *Razón Social* (B2B); nationality/country; phone and email; for B2B a contact person (first name, last name, job title); credentials managed by the authentication system.
Employee	Document type (DNI, CE, PAS) and number; first/last name; nationality/country; gender; profile picture; birthdate; hire date; active/inactive; role; personal and business phone/email; credentials.
Vehicle model	Name; description; one or more categories (for example *Camioneta, Comercial, Familiar, Hatchback, Sedan*); available versions and colors.
Version	Name; reference price; active/inactive (or equivalent archival).
Color and images	Colors vary per model; per-color images for six viewpoints: main (required), exterior front, left, right, rear, and interior.

Entity	Description and key information
Sale register	Customer reference and snapshot; vehicle selection and price snapshot; 13 task records; current active task; temperature (`Hot`, `Warm`, `Cold`); overall state; notes; assigned advisor; provenance fields.
Task instance	One of the 13 ordered tasks per sale with status, task-specific data, optional notes, timing and completion information.
Document	Uploaded file with type/MIME, size in bytes, original name, upload timestamp, uploader, task/sale association, visibility and version history.
Notes	Sale-level notes and per-task notes.
Audit event	Append-only record of significant changes (Section 7.1).
Access log	Append-only authentication log, separate from the audit trail (Section 6).
Notification and log	Customer and employee notifications, delivery attempts and outcomes, reminder state (Section 7.2).

3.2 Customer categories and identification

Group	Customer types	Identification documents
B2C	Diplomatic customer	DNI, CE, PAS, CDI
B2B	Embassy; Catholic religious entity; non-Catholic religious entity; International Organization	RUC

Document type plus document number must uniquely identify a customer, and likewise an employee, within its own table. Document numbers are identifiers, not numbers: leading zeros must be preserved. **[Confirmed]** For Task 2, the *signer* is the customer for a Diplomatic customer and the representative/executing person for an Embassy, Religious Entity or International Organization. **[Confirmed]**

3.3 Sale register rules

- Each sale register represents exactly one vehicle sale; multiple purchases create independent sale registers. **[Confirmed]**
- All 13 task instances are created with the sale, initially `pending`. **[Confirmed]**
- Overall sale state is `active`, `completed` or `cancelled`. A sale becomes `completed` automatically when all 13 tasks are completed. **[Confirmed]**
- Sales advisors may cancel their own sales with a reason but cannot undo a cancellation. Sales managers may cancel any sale and undo cancellations. **[Confirmed]**
- Cancelled sales are retained for history and audit and are not physically deleted in ordinary operation; System Admins may permanently delete corrupted or test sale records under safeguards and audit. **[Confirmed]**
- The creator (`created_by`) is recorded separately from the assigned Sales Advisor, who may change by reassignment. **[Confirmed]**
- **[Open]** Who sets the sale temperature (`Hot`/`Warm`/`Cold`), and whether it changes over time.
- **[Open]** Behavior of tasks and timers when a sale is cancelled and when a cancellation is undone; and whether reverting Task 13 returns a completed sale to `active` (proposal: yes,

transactionally, with an audit event).

3.4 Historical snapshot rules

Snapshot	Rule
Customer snapshot	At sale creation, capture the customer's legal/name value, document type and document number. Sale records and snapshots are immutable for financial and legal audit, even if the customer profile changes later. **[Confirmed]**
Vehicle price snapshot	At sale creation, capture the selected version's reference price; later catalog price changes must not alter the sale. **[Confirmed]**
Vehicle descriptors	The sale references the model/version/color. **[Open]** Whether to also snapshot display names so later catalog renames do not change historical sales (proposal: snapshot names as well).

3.5 Record provenance

For customers, employees, vehicle catalog records (models, versions, colors, images) and sale registers, the platform tracks `created_at`, `created_by`, `updated_at`, `updated_by` where applicable, and archive/deactivation metadata for entities that are archived rather than deleted. **[Confirmed]** Employee records identify the System Admin who created them, and catalog changes identify the employee who made them. System-seeded, imported or bootstrap records use an explicit system/migration actor or a documented nullable-field policy rather than an invented human creator. **[Confirmed]** Provenance fields do not replace the audit trail (Section 7.1).

4 The 13-Task Operational Lifecycle

4.1 Status model and strict linear transition rules

Each task has exactly one of three statuses: `pending`, `in_progress`, `completed`. **[Confirmed]**

- The process is strictly linear. Task 1 is the only task eligible to start initially; Task N may move from `pending` to `in_progress` only after Task $N-1$ is `completed`. **[Confirmed]**
- Parallel tasks, branches and out-of-order completion are not allowed in the normal workflow; at most one task is `in_progress` per sale. **[Confirmed]**
- The current active task is the single `in_progress` task; if none exists, the next eligible task is derived from the ordered statuses. **[Confirmed]**
- No dependency table or graph is introduced for the linear process unless a future requirement justifies branching. **[Confirmed]**
- A confirmation message is shown before any task status change, especially completion. **[Confirmed]**
- All backend endpoints, not only UI controls, enforce sequencing, required data, uniqueness and permissions. **[Confirmed]**

Transition	Allowed actors	Preconditions	Notes
`pending` → `in_progress`	Advisor (own sale), Manager, Admin	Task $N-1$ is `completed` (Task 1: sale is `active`)	Records start timestamp; due date is computed here (Section 7.3); confirmation required.
`in_progress` → `completed`	Advisor (own sale), Manager, Admin	All required fields and documents present; validations pass	Records completion timestamp and `completion_phase`; may trigger a notification; confirmation required.
`completed` → `in_progress`	Manager, Admin only	Task is the *most recently completed* task; reason recorded	No cascading rollback; fully audited.
`in_progress` → `pending`	**[Open]**	**[Open]**	Not specified in the brief; see Appendix A.

4.2 Task data and editing rules

- Every task has an optional Notes textarea. **[Confirmed]**
- Every non-optional field or document is required before the task can be completed. **[Confirmed]**
- All task fields and document uploads are disabled and read-only until the task becomes `in_progress`. **[Confirmed]**
- Unique constraints specified per task must be enforced. **[Confirmed]**
- After completion, only a Sales Manager or System Admin may modify fields or documents; Sales Advisors see completed tasks read-only. **[Confirmed]**
- **Data correction:** Managers and Admins may edit fields or replace documents of a `completed` task without changing its status; every such edit requires a mandatory "Reason for Edit" and is recorded in the append-only audit log. **[Confirmed]**
- **Accidental completion:** only the most recently completed task may be reverted to `in_progress`. To reopen Task 4 while Task 6 is active, a manager must revert Task 6, then Task 5, then Task 4. This friction is intentional to protect data integrity and customer milestone visibility. **[Confirmed]**
- The recommended duration is the elapsed time from `in_progress` to `completed`; tasks without a recommended duration have no predefined duration. **[Confirmed]**

4.3 Financial reconciliation rules

All amounts are USD. Let Q be the *quoted final vehicle price* of Task 1, T_0 and C_0 the initial Toyota amount and commission, A the accessory amount, R the vehicle release processing amount, and T_1, C_1 the final Toyota amount and commission (Task 3).

Rule	Definition
F1	Task 1: $Q = T_0 + C_0 + A + R$ (calculated). T_0 and C_0 are provisional; A and R stay constant for the whole operation. **[Confirmed]**
F2	Task 3: the quoted price does not change. $C_1 = Q - (T_1 + A + R)$, so that $T_1 + C_1 + A + R = Q$. **[Confirmed]**

Rule	Definition
F3	Task 4: the deposited amount is expected to equal Q. **[Confirmed]** **[Open]** Whether a mismatch blocks completion or requires a documented reason.
F4	Task 5: the deposited amount is expected to equal T_1. **[Confirmed]**
F5	Task 6: invoice amounts are expected to equal T_1 (Toyota), C_1 (commission), A (accessories) and R (release processing). **[Confirmed]** The four invoices therefore sum to Q.
F6	**[Proposal]** Store money as exact decimals (for example two decimal places), never floating point; define a rounding rule. **[Open]** If C_1 would be negative (that is, $T_1 + A + R > Q$), whether to block, or require manager approval and a reason.

4.4 Task summary

#	Task	Duration	Milestone	Customer email on completion
1	Cotización del Vehículo	None	1	Yes (quotation ready)
2	Debida Diligencia del Cliente	3 days	1	No
3	Liquidación de Unidad	12 days	2	Yes (vehicle assigned)
4	Pago a Grupo Pana	None	2	No
5	Pago a Toyota	None	3	No
6	Facturación	None	3	Yes (invoices issued)
7	Documentos Endosados	10 days	4	Yes (customs begins)
8	Trámite Aduanero	12 days	4	No
9	Trámite de Liberación	30 days	5	No
10	Inmatriculación y Placas	15 days	5	Yes (plate assigned)
11	Traslado a PDS	None	6	Yes (final preparation)
12	Equipamiento	None	6	No
13	Entrega de Vehículo	None	6	Yes (delivery completed)

4.5 Task specifications

Fields marked [CV] are customer-visible. All uploaded documents record at least MIME type and size in bytes. **[Confirmed]**

Task 1 — Cotización del Vehículo (duration: none)

Purpose: prepare the official quotation, establishing the quoted total and its initial breakdown.

- Initial amount payable to Toyota, USD (numeric)[CV] — provisional, finalized in Task 3
- Initial Grupo Pana commission, USD (numeric)[CV] — provisional, finalized in Task 3

- Accessory amount, USD (numeric)[CV] — constant for the operation
- Vehicle release processing amount, USD (numeric)[CV] — constant for the operation
- Quoted final vehicle price, USD (calculated sum of the four amounts)[CV]
- Official vehicle quotation — PDF[CV]
- Official vehicle quotation — Excel (not customer-visible)

Task 2 — Debida Diligencia del Cliente (duration: 3 days)

Purpose: collect the signer's identity documents and the Customer Knowledge Affidavit, determine PEP status and record the DDC regime.

- Signer's document type (DNI, CE, PAS, CDI)[CV]; document number[CV]; first name[CV]; last name[CV]
- Scanned identity document — PDF[CV]
- Customer Knowledge Affidavit — PDF[CV]
- Power of Attorney validity document (*Vigencia de Poder*) — PDF[CV]; required if the customer type is not *Diplomático*
- PEP status: `No PEP`, `PEP`, `Relacionado con PEP` (not customer-visible)
- DDC regime: `General`, `Reforzado` (not customer-visible)

Task 3 — Liquidación de Unidad (duration: 12 days)

Purpose: assign the vehicle's series and VIN, finalize the Toyota amount and commission, and upload the unit settlement.

- Vehicle series number (unique)[CV]
- Vehicle Identification Number, VIN (unique)[CV]
- Final amount payable to Toyota, USD[CV]
- Final Grupo Pana commission, USD (calculated per rule F2)[CV]
- Settlement document — PDF (not customer-visible)

Task 4 — Pago a Grupo Pana (duration: none)

Purpose: verify the customer's payment.

- Transaction bank (BCP, BBVA, Scotiabank, Interbank and other supported banks)[CV]
- Transaction date[CV]; transaction number (unique)[CV]
- Deposited amount, USD (expected to equal Q)[CV]
- Payment voucher — PDF[CV]

Task 5 — Pago a Toyota (duration: none)

Purpose: record the internal payment to Toyota. No fields are customer-visible.

- Transaction bank; transaction date; transaction number (unique)

- Deposited amount, USD (expected to equal T_1)
- Payment voucher — PDF; payment confirmation — PDF

Task 6 — Facturación (duration: none)

Purpose: record the four official invoices. For each invoice all four fields are customer-visible[CV].

- Toyota invoice: number (unique), date, amount (expected T_1), PDF
- Commission invoice: number (unique), date, amount (expected C_1), PDF
- Accessories invoice: number (unique), date, amount (expected A), PDF
- Vehicle release processing invoice: number (unique), date, amount (expected R), PDF

Task 7 — Documentos Endosados (duration: 10 days)

Purpose: receive and upload the endorsed documents needed for customs.

- Endorsed documents — PDF (not customer-visible)

Task 8 — Trámite Aduanero (duration: 12 days)

Purpose: manage nationalization/customs processing. No fields are customer-visible.

- Deposit order (*Pedido de Depósito*) — PDF
- Customs Declaration of Goods (DAM) number
- Customs office (dropdown of customs offices in Peru) **[Open]** source of the list
- Control channel: `Verde`, `Naranja`, `Rojo`
- Customs certificate — PDF

Task 9 — Trámite de Liberación (duration: 30 days)

Purpose: assign an internal person in charge and track the release period.

- Person in charge (textbox; not customer-visible)

Task 10 — Inmatriculación y Placas (duration: 15 days)

Purpose: register the vehicle, assign the definitive plate and upload the TIVe.

- License plate number (unique)[CV]
- Registration date[CV]
- Electronic Vehicle Identification Card (TIVe) — PDF[CV]

Task 11 — Traslado a PDS (duration: none)

- Transfer date[CV]

Task 12 — Equipamiento (duration: none)

- Service order number; service order — PDF (not customer-visible)
- Equipment completion date[CV]

Task 13 — Entrega de Vehículo (duration: none)

- Transfer date to Surquillo (optional)[CV]
- Vehicle delivery date[CV]
- Signed vehicle delivery record — PDF[CV]

4.6 Validation rules

The brief requires validations for amounts, dates, identifiers and required documents. The concrete rules below are proposals for confirmation. **[Proposal]**

- Amounts: non-negative exact decimals in USD; calculated fields are computed server-side and cannot be overwritten.
- Dates: valid calendar dates; transaction, invoice and registration dates are not in the future (to be confirmed).
- Identifiers: unique constraints for series, VIN, transaction numbers, invoice numbers and plate. **[Open]** Scope of uniqueness: for example transaction numbers per bank, invoice numbers per series, and whether series/VIN/plate of a *cancelled* sale may be reused.
- Documents: allowed formats PDF (and Excel for the Task 1 quotation); maximum size and content-type verification to be defined (see Section 7.4).

5 Customer Experience and the Six Macro-Milestones

5.1 Customer portal capabilities

Customers can sign in through a customer-specific interface, view all sale registers associated with their customer record, view vehicle information and the assigned advisor (full name, contact email and phone), follow progress through six milestones, access only customer-visible fields and documents, receive the defined emails and reset their password. **[Confirmed]** Customers must not be able to view or infer another customer's sales, documents or private information. **[Confirmed]**

For each sale the customer sees: vehicle model, version, categories, reference-price snapshot, selected color and images; assigned advisor contact details; sale creation timestamp, overall state and progress timeline. **[Confirmed]**

5.2 Customer account lifecycle

- Employees create customer accounts, subject to permissions. **[Confirmed]**
- The system generates a secure temporary password and emails the portal link and login data (document type, document number, temporary password). **[Confirmed]**
- On first login with the temporary password the customer is forced to a mandatory "Set New Password" screen and cannot use the portal until a new secure password is saved. **[Confirmed]**

- Customers cannot edit their document, name, email or phone through the portal; they can reset their password through "Forgot Password". **[Confirmed]**
- **[Proposal]** Because the email carries a credential, make the temporary password single-use with a short expiry, or replace it with a one-time activation link, while keeping the forced-change behavior.

5.3 Milestones

#	Milestone	Tasks	Meaning for the customer
1	Cotización y Documentación	1, 2	Official quotation established and signer's identity documentation validated.
2	Asignación y Pago	3, 4	Physical vehicle assigned by series number; payment to Grupo Pana enabled.
3	Facturación Oficial	5, 6	Toyota payment processed and official invoices issued.
4	Trámites Aduaneros	7, 8	Endorsed documents processed; vehicle in customs/nationalization.
5	Liberación y Placas	9, 10	Release period and progress toward registration and definitive plate.
6	Preparación y Entrega	11, 12, 13	Move to pre-delivery inspection, equipment installation and final delivery coordination.

5.4 Derived milestone state

- A milestone is `completed` when all its tasks are completed; `in_progress` when its task group contains the single active task; otherwise `pending`. **[Confirmed]**
- The current milestone is **not stored** as an editable field on the sale register; it is derived from ordered task statuses and exposed consistently through the API, operations filters and BI queries (for example through a database view or reusable query). **[Confirmed]**
- A cached or denormalized value is allowed only if measurements show a real performance problem, and then it must be updated transactionally and never become a second source of truth. **[Confirmed]**
- Future milestones cannot be active or completed ahead of earlier ones. The overall sale state (`active`, `completed`, `cancelled`) is displayed separately from milestone state. **[Confirmed]**
- **[Assumption]** Because a manager may revert the most recently completed task, a customer may occasionally see a milestone move backward; the portal should present this neutrally.

5.5 Customer-visible data

Customer visibility is enforced **server-side**; a frontend flag is not an authorization mechanism. **[Confirmed]** Task fields marked ^{CV} in Section 4 are the only task data exposed. Not customer-visible: the Task 1 Excel quotation, PEP status, DDC regime, the Task 3 settlement document, all of Tasks 5, 7, 8 and 9, and the service order of Task 12. **[Assumption]** Notes are internal unless a future rule designates otherwise.

5.6 Customer email notifications

Emails are tied to high-value events and required actions rather than every internal transition. **[Confirmed]**

#	Trigger	Subject (Spanish)	Content summary
1	Advisor registers the customer for the first time (once per customer)	Bienvenido a Grupo Pana - Acceso a tu portal de seguimiento	Welcome, portal link, login identifier and temporary password; must state the password has to be changed at first login.
2	Advisor creates the sale register and assigns the vehicle	Hemos registrado tu nueva solicitud de vehículo	Process started; vehicle model/color; advisor name and contact number; link to view the first milestone.
3	Task 1 completed	Tu cotización oficial está lista para revisión	Quotation available; link to download the PDF from the authenticated portal.
4	Task 3 completed	Siguiente paso: Tu vehículo ha sido asignado	Vehicle assigned with series number and VIN; direct to portal for payment to Grupo Pana.
5	Task 6 completed	Tus comprobantes de pago han sido emitidos	Payment/invoicing steps recorded; invoices available for download.
6	Task 7 completed	Iniciamos los trámites aduaneros de tu vehículo	Endorsed documents processed; customs underway; no unsupported delivery-date promise.
7	Task 10 completed	ąTu vehículo ya cuenta con placa oficial!	Definitive plate and secure link to the TIVe PDF.
8	Task 11 completed	Tu vehículo está en fase de preparación final	Vehicle entered pre-delivery preparation/equipment phase.
9	Task 13 completed	ąFelicidades por tu nuevo vehículo! / Gracias por confiar en Grupo Pana	Thank you, delivery date, secure link to the signed delivery record.

Notification 2 refers to “Phase 1”; for consistency with the customer model this should read “Milestone 1”. **[Proposal]**

6 Security, Authentication and Role-Based Access Control

6.1 Actors

Actor	Description
System Admin	Full administrative control, including staff management and exceptional destructive operations under safeguards.
Sales Manager	Global business and operational control over sales, overrides and reassignment.
Sales Advisor	Manages own customers and sales and works own active tasks.
Customer	Separate access class (not in the employee RBAC table): own records and customer-visible content only.
System actor	[Proposal] Non-human actor for scheduler, migrations and seeded data, used for provenance and audit.

6.2 Role-permission matrix

Module	System Admin	Sales Manager	Sales Advisor
Staff management	Full CRUD: create staff, assign roles, reset passwords, permanently delete when genuinely necessary	View active staff, limited to Sales Advisors, for reassignment	No access
Vehicle catalog	Full CRUD	Create/edit models, upload photos, archive models/versions	View only; browse/select for a sale
Customers	Full CRUD, including exceptional permanent deletion	View/edit/create any customer; no hard delete	Create customers; view/edit only customers associated with own sales
Sale registers	Full CRUD; delete corrupted/test records under controlled conditions	View all, reassign advisors, cancel any sale, undo cancellations	Create/update/cancel own sales (reason required); cannot undo cancellation
Tasks and documents	Full access for exceptional intervention	Override tasks across all sales, edit/replace files, reopen completed tasks	Update own active tasks; completed tasks read-only
BI dashboard	Global view	Global business/operational view	Individual view of own pipeline and permitted metrics
Raw exports	Full-system export including audit/access logs	Business exports: customer list, sale registers, task audit logs, cancellation logs	No raw exports

[Confirmed] The matrix above is the intended permission model.

Permission design requirements

- Authorization is enforced in the backend/API, with least privilege by default. [Confirmed]
- Viewing is separated from modifying, cancelling, reopening, exporting and deleting. [Confirmed]

- Security-sensitive actions, overrides, cancellations, role changes and exceptional deletions are logged. [Confirmed]
- Prefer soft deletion/archival; permanent removal requires confirmation and an audit trail. [Confirmed]

Concerns and proposed resolutions

- [Open] **Advisor customer access.** Advisors may create customers but view only customers tied to their own sales; a newly created customer has no sale yet, and a uniqueness conflict on document number reveals that a customer exists. [Proposal] Treat customers created by the advisor as accessible, and offer a controlled "attach existing customer" flow that exposes minimal data.
- [Open] **Reassignment.** Whether access changes immediately when a sale is reassigned (proposal: yes, access follows the current assignee; history stays in the audit trail).
- [Proposal] **Admin destructive power.** Require a reason, an explicit confirmation and an audit event for permanent deletion; consider a restricted time window or two-person approval later.

6.3 Authentication

- **Two interfaces and routes:** employees at `/admin` (document types DNI, CE, PAS) and customers at `/portal` (DNI, CE, RUC, PAS, CDI). [Confirmed] Separate pages are for usability and routing only; identity, user type, role and resource authorization are enforced by the backend. [Confirmed]
- **Login fields (both):** document type, document number, password, a "Forgot password" link and a "Keep me signed in" checkbox. Email is not the login identifier. [Confirmed]
- **Validation:** document format depends on the selected type (for example DNI typically eight digits, RUC eleven), configurable and validated on both client and server. [Confirmed]
- **Uniqueness:** composite unique `(document_type, document_number)` in both the customers and employees tables, normalized consistently while preserving leading zeros. [Confirmed]
- **Same person in both tables:** the authentication system must distinguish account types and prevent cross-portal leaks. [Proposal] Separate account namespaces, a mandatory `user_type` claim and token audience, and separate cookie names/paths.
- **Sessions:** short-lived JWT access tokens and long-lived refresh tokens in HTTP-only cookies, with a Remember Me option. [Confirmed] [Proposal] Add `Secure` and `SameSite` flags, refresh-token rotation and revocation, CSRF protection where applicable, logout invalidation and session expiry; JWT alone is not a complete strategy.
- **Passwords:** never stored in plain text; use a modern password-hashing algorithm. [Confirmed]
- **Access logs:** an append-only `AccessLogs` table records `user_type`, attempted document type/number, `user_id` when identified, `action_type` (`LOGIN_SUCCESS`, `LOGIN_FAILED`, `LOGOUT`), IP address where reliable, and timestamp. No secrets or passwords are logged. [Confirmed]

6.4 Security controls

The brief requires evaluation of at least the following; the baseline controls are proposals. **[Proposal]**

Area	Baseline control
Customer isolation	Every customer query is scoped by the authenticated customer id on the server; opaque identifiers; tests that attempt cross-customer access (IDOR).
Document access	Private storage; downloads only through backend authorization with short-lived links; no public URLs for private documents.
Brute force	Rate limiting and progressive lockout on login and password-reset endpoints; all failures logged in `AccessLogs`.
Input and uploads	Server-side validation with a schema library; file type, size and content verification.
Sensitive data exposure	Exclude passwords, tokens and document contents from logs, audit payloads, analytics and API responses.
Web vulnerabilities	Standard protections: parameterized queries, output encoding, security headers, CORS allow-list, dependency scanning.
Audit integrity	Append-only tables for ordinary roles; restricted access and defined retention.
Privacy and legal	**[Open]** Review obligations under Peruvian personal-data protection law with qualified counsel, including consent, retention and handling of identity and due-diligence documents.

7 Data Provenance, Event Auditing and Notification Engine Rules

7.1 Append-only audit trail

The audit trail is distinct from provenance fields and from authentication access logs. It is append-only for ordinary roles, and it avoids storing passwords, tokens, full document contents or unnecessary sensitive data. **[Confirmed]** Each event records the actor, target entity and record id, event/action type, timestamp, and an appropriate before/after summary. **[Confirmed]**

Event class	Minimum coverage
Task status changes	Previous/new status, actor, timestamp, reason where applicable.
Sale state changes	Cancellation, reopening, completion changes, and the reason for cancellation/override.
Assignment	Assignment and reassignment of Sales Advisors.
Field changes	Important changes to sale, customer, vehicle/catalog and task fields; edits to completed tasks with the mandatory “Reason for Edit”.
Documents	Uploads, replacements, archival/deletion and visibility changes.
Security	Role/permission changes and other security-sensitive administrative actions, including permanent deletions.

[Open] Access restrictions and retention periods for audit logs and access logs.

7.2 Notification engine rules

- Notifications must be reliable and idempotent: no duplicates if a task transition is retried, a scheduler runs twice, or an event is replayed. **[Confirmed]**
- Delivery attempts and outcomes are stored. **[Confirmed]**
- Sensitive documents are not attached to emails; secure authenticated links are used instead. **[Confirmed]**
- **[Proposal]** Use a transactional outbox: the notification is written in the same database transaction as the task transition, then delivered asynchronously with retries; a unique key per sale, event type and recipient provides deduplication.
- **[Open]** If a task is reverted and completed again, whether the customer email is resent (proposal: no resend; the portal shows the corrected state).

7.3 Task deadlines, reminders and escalation

- **Due date:** computed when a task goes from `pending` to `in_progress` by adding the recommended duration to the start timestamp, in strict elapsed calendar time (weekends and non-business hours do not pause the clock). Example: a three-day task started Friday 4:00 PM is due Monday 4:00 PM. **[Confirmed]**
- Tasks without a recommended duration receive no duration-based reminders. **[Confirmed]**
- **Approaching Due:** one warning when 80% of the recommended duration has elapsed (for the three-day task, after 57.6 hours). **[Confirmed]**
- The platform never automatically unlocks or reassigns a task because it is overdue. **[Confirmed]**

Level	Recipient	Additional behavior
Level 1: warning through 24 hours overdue	Assigned Sales Advisor	Visible indicator.
Level 2: 24 to 48 hours overdue	Sales Manager	Escalation indicator.
Level 3: more than 48 hours overdue	System Admin	Task visually highlighted in the UI.

[Confirmed] **[Open]** Whether "Sales Manager" and "System Admin" mean all holders of the role or a designated person.

Recurring reminders. Frequency depends on the recommended duration (shorter tasks remind more often) with a hard cap; the exact intervals and cap are to be confirmed before implementation. **[Confirmed]** **[Proposal]** Initial proposal: tasks of at most 3 days every 12 hours; tasks of 10–15 days every 24 hours; the 30-day task every 48 hours; at most 6 recurring reminders per task.

Deduplication state and logging. Track `last_reminder_sent_at`, `current_reminder_phase` (`approaching`, `overdue`, `recurring`) and `recurring_reminder_count`; record every dispatched reminder in an append-only `Notification_Log` (or equivalent). **[Confirmed]**

UI and analytics. Show indicators for *Approaching*, *Overdue* and *Escalated/Recurring*, with quick filters for urgent tasks. **[Confirmed]** Store `completion_phase` on completion with values `early`, `in_approaching`, `overdue` or `recurring_overdue`. **[Confirmed]** **[Open]** Value of `completion_phase` for tasks without a recommended duration (proposal: null).

7.4 File and document management

The platform must support secure upload, validation and storage; MIME type and size in bytes per upload; original file name, upload timestamp, uploader and task/sale association; replacement behavior with version history where needed; deletion/deactivation rules with audit history; defined maximum size and formats; role-based upload, update and download permissions; explicit per-field/document customer visibility; authenticated or time-limited access; and safe behavior when a stored file is missing. **[Confirmed]** The database stores metadata and storage references rather than file contents. **[Confirmed]** Private documents must never be exposed merely because a URL is known. **[Confirmed]** **[Open]** Maximum file size and allowed formats (proposal: PDF, plus Excel for the Task 1 quotation; a size limit to be set from storage quotas).

8 Proposed System Architecture and Technology Selection

*Everything in this section is a technical recommendation **[Proposal]**; the brief states that its stack is an initial preference to be evaluated under the zero-cost constraint. Architecture and schema are finalized only after rules, statuses, permissions, file flows and reporting are understood.*

8.1 Architectural principles

- Prefer the simplest architecture that meets the requirements: a **modular monolith** with a React frontend, an Express/TypeScript API and PostgreSQL; no microservices.
- Keep business rules (task sequencing, snapshots, permissions, visibility) in the backend domain layer, independent of HTTP and the database.
- Use clean-code practices, SOLID principles and safe design patterns; provide a complete modular project structure before generating individual files. **[Confirmed]**

8.2 Logical architecture

```
Browser (React + TypeScript)
  |-- Employee App (/admin)   |-- Customer Portal (/portal)
  v  HTTPS, JSON, HttpOnly cookies
API (Node.js + Express + TypeScript, modular monolith)
  |-- Auth & RBAC          |-- Customers / Employees / Catalog
  |-- Sales & Tasks        |-- Documents (storage abstraction)
  |-- Notifications        |-- Analytics (SQL views/queries)
  |-- Internal job endpoint (scheduler tick)
  v                        v                      v
PostgreSQL             Private file storage     Email provider
(data, audit,          (signed, short-lived     (transactional,
 outbox, logs)          access via API)          delivery log)
```

8.3 Technology selection under the zero-cost constraint

Provider information was checked on 15 October 2026 from public documentation and third-party summaries; **sources disagree on some details, and quotas must be re-verified on each provider's official page before relying on them.**

Component	Proposal	Free-tier findings and risks
Frontend	React + TypeScript (Vite)	Static hosting is free on Render and comparable hosts.
Backend	Node.js, Express, TypeScript	A free Render web service spins down after about 15 minutes without traffic (roughly one minute to wake) and shares a monthly instance-hour allowance; documentation advises against production use. In-process cron is therefore unreliable.
Database	PostgreSQL	Render's free Postgres is time-limited (reported as 30 days plus a grace period, with some sources saying 90 days), so it only suits brief demos. A free Postgres such as Supabase (reported 500 MB per project, two projects, pause after about one week idle) is a candidate used as plain PostgreSQL to keep migration easy. **[Open]**
Scheduler	Idempotent `tick` endpoint triggered externally (for example a scheduled GitHub Actions workflow)	Wakes the sleeping API and exercises the database; must tolerate delays and duplicate runs (hence deduplication state). **[Open]**
File storage	Storage abstraction; Cloudinary for catalog images; private bucket for confidential documents	Cloudinary free plan reports 25 credits per month (1 credit ≈ 1 GB storage or bandwidth, or 1,000 transformations) and three users. Private/authenticated delivery with signed URLs is available, but token-based expiring URLs are reported as a paid feature and exceeding credits can suspend asset serving. For identity, due-diligence, financial and legal documents, a private bucket with backend-authorized short-lived links (for example a free-tier object store) is the safer default. **[Open]**
Authentication	Own JWT + rotating refresh cookie	No third-party cost or lock-in; see Section 6.
Email	Provider with a free tier, to be evaluated	Quotas not yet verified.
BI	SQL views and parameterized queries with a chart library	The brief asks for the simplest maintainable approach; no separate analytics platform.

8.4 Modeling task-specific data

The brief requires comparing approaches rather than forcing all fields into one unstructured column. **[Confirmed]**

Option	Advantages	Drawbacks	Fit
Fully relational (one table per task type)	Strong typing, native unique and foreign-key constraints, easy reporting	More tables and migrations; changes to a task need schema changes	Strong for constrained and reported fields

Option	Advantages	Drawbacks	Fit
Single JSON column	Flexible, few tables	Weak constraints (uniqueness, amounts), harder reporting, validation only in code	Risky for this domain
Hybrid	Typed columns/tables where integrity or reporting matters; JSON only for informational data	Two patterns to maintain	Leading candidate

[Proposal] Provisional direction: a hybrid in which values that need uniqueness, arithmetic reconciliation or reporting (amounts, series, VIN, transaction and invoice numbers, plate, dates, enumerations) are typed relational columns, with a common task-instance table, a generic document table and an ordered task-definition table. This will be settled in an ADR during data modeling.

8.5 Backend and frontend structure

[Proposal] Modular monolith with layered, feature-oriented modules (presentation, application, domain, infrastructure).

```
endotrack/
  apps/
    api/                        # Express + TypeScript
      src/
        modules/
          auth/ customers/ employees/ catalog/
          sales/ tasks/ documents/ notifications/ analytics/
            (each: http/ application/ domain/ infrastructure/)
        shared/                 # errors, validation, logging, config
        jobs/                   # scheduler tick, outbox dispatcher
      tests/
    web/                        # React + TypeScript
      src/
        features/               # per-module UI, hooks, API clients
        components/ routes/ lib/
  packages/
    contracts/                  # shared Zod schemas and types
  docs/ (adr, erd, api)
```

Candidate patterns: repository and unit-of-work for persistence, a task state machine in the domain layer, policy objects for authorization, outbox for notifications, and a strategy/adapter for storage and email providers. **[Proposal]**

8.6 Key technical risks

- Free-tier behavior (sleeping API, expiring or pausing databases, storage quotas) conflicts with scheduling and confidential-file requirements; mitigated by the scheduler design and storage abstraction above.

- Strict-sequence integrity under concurrent updates requires transactional checks and database-level constraints in addition to application logic.
- Confidential identity and financial documents require a storage and retention policy reviewed for legal compliance.

A Open Decisions Register

ID	Question	Proposed resolution
D1	Allow `in_progress` → `pending` (task started by mistake)?	Not allowed; use correction by a manager if ever needed.
D2	Behavior of timers and tasks when a sale is cancelled or restored.	Freeze timers on cancellation; resume on restore with audit.
D3	Does reverting Task 13 reopen a completed sale?	Yes, transactionally with an audit event.
D4	Scope of uniqueness: series, VIN, plate (cancelled sales), transaction numbers (per bank), invoice numbers (per series).	Unique among non-cancelled sales; transactions unique per bank.
D5	Mismatch handling for expected amounts (F3–F5) and negative final commission.	Soft warning plus mandatory reason; manager approval for negative commission.
D6	Advisor customer visibility and "attach existing customer" flow.	Controlled flow exposing minimal data.
D7	Does reassignment change access immediately?	Yes, access follows the current assignee.
D8	Recurring reminder intervals and cap; meaning of Manager/Admin recipients.	See Section 7.3; confirm before implementation.
D9	Required image viewpoints beyond the main view.	Configurable; main view required.
D10	Maximum file size and allowed formats; retention of documents, audit and access logs.	To be set from storage quotas and legal review.
D11	Storage provider for confidential documents.	Private bucket with signed short-lived access; Cloudinary for catalog images only.
D12	DNI/RUC name-lookup provider (availability, terms, privacy, rate limits, legal suitability).	Evaluate before choosing; manual entry always available; no scraping without review.
D13	Who sets sale temperature; definition of progression/conversion rate for BI.	Advisor sets temperature; define KPIs in the analytics specification.
D14	Resend of customer emails after a task is reverted and re-completed.	No resend.
D15	UI languages and the source of the Peruvian customs-office list.	To be confirmed with the business.
D16	Customer-account email credential design.	Single-use, expiring temporary password or activation link.

B Revision History

Version	Date	Description
0.1	15 Oct 2026	Baseline specification derived from the product brief; open decisions registered.